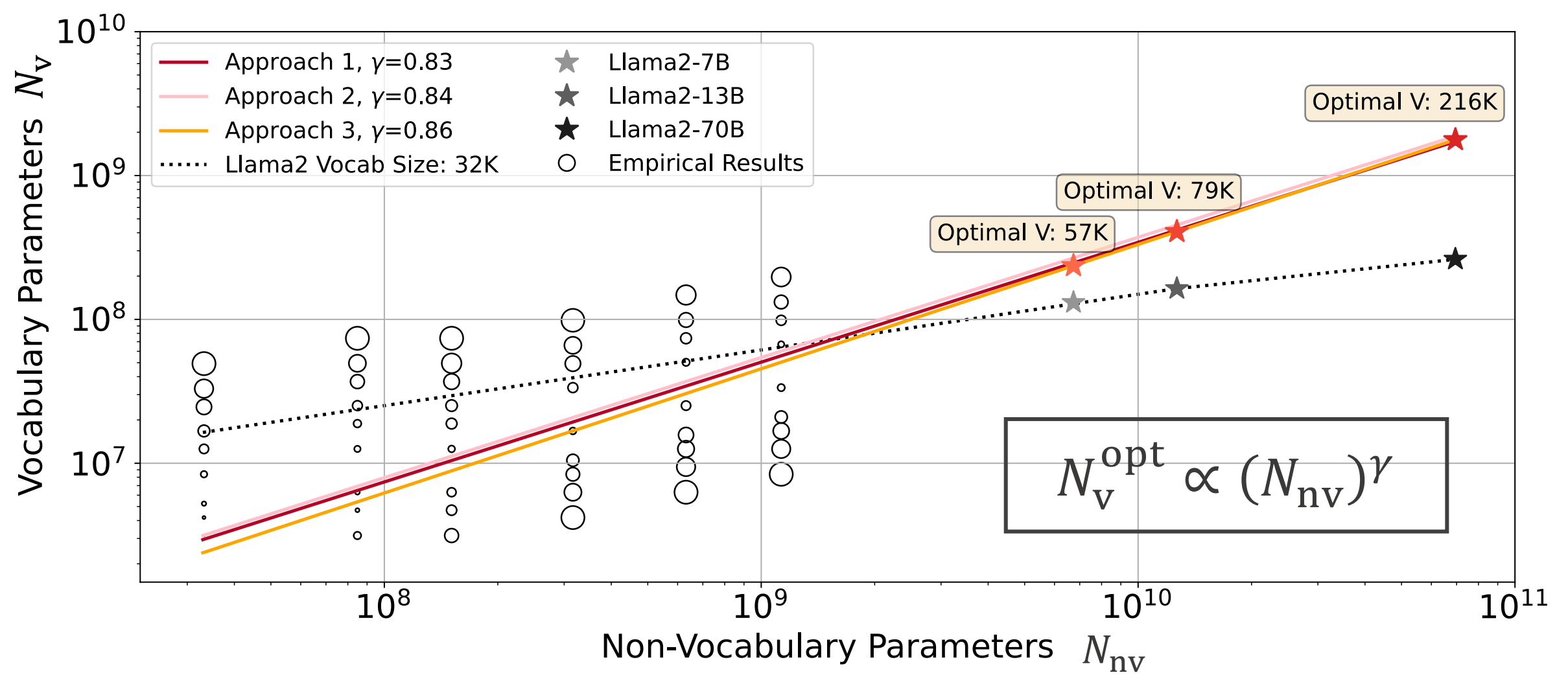
Approach 1, γ=0.83
Approach 2, γ=0.84
Approach 3, γ=0.86
Llama2 Vocab Size: 32K
Llama2-7B
Llama2-13B
Llama2-70B
Empirical Results
Optimal V: 216K
Optimal V: 79K
Optimal V: 57K
$N_{\mathrm{v}}^{\mathrm{opt}} \propto (N_{\mathrm{nv}})^{\gamma}$
Vocabulary Parameters N_{v}
Non-Vocabulary Parameters N_{nv}
10^{7}
10^{8}
10^{9}
10^{10}
10^{11}